Field Effect Transistor (FET)

Introduction

- Field effect Transistor is a semiconductor device whose operation depends on the control of current by an electric field.
- The current conduction in FET is due to majority charge carriers only. Therefore, FETs are known as unipolar devices.
- BJT is current control device and FET is a voltage control device.

Introduction

FET has several advantages over BJT

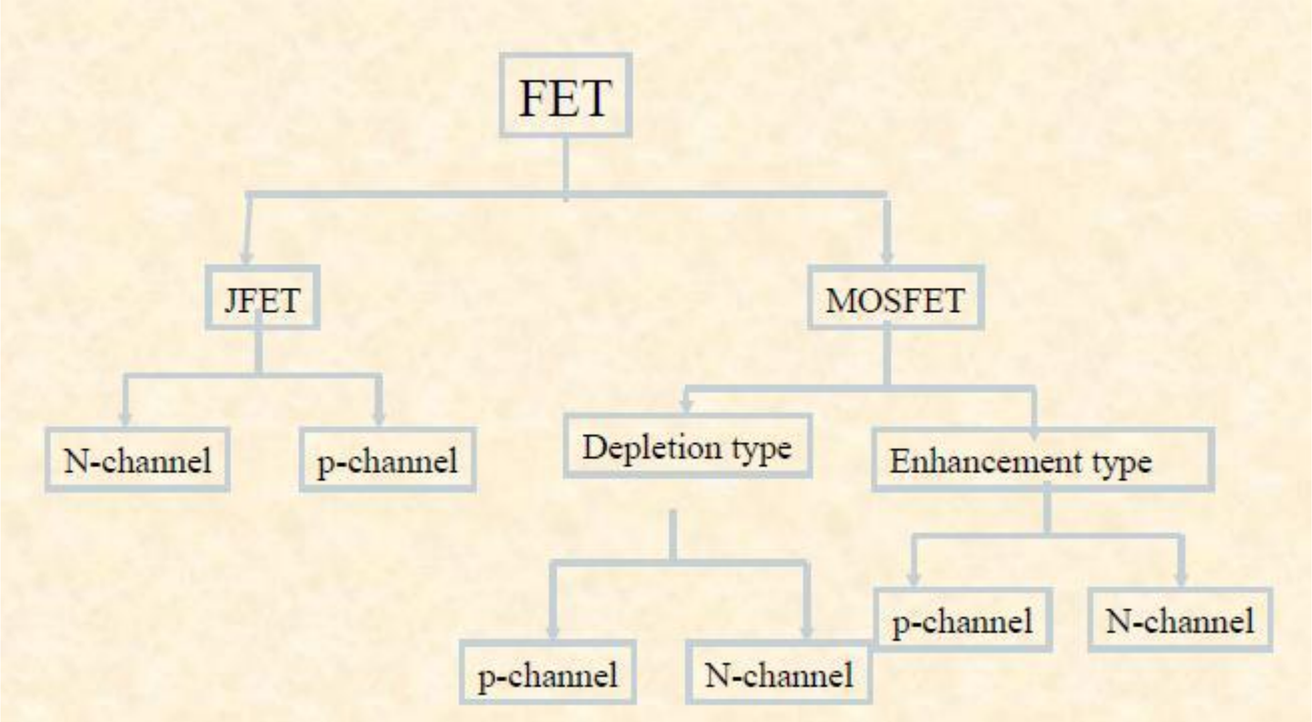
- Current flow is due to majority carriers only
- Immune to radiation
- High input resistance
- Less noisy than BJT
- No offset voltages at zero drain current
- High thermal stability

Disadvantages:

- FET is less sensitive to the change in applied voltage than BJT. Therefore, voltage gain of FET is less than BJT.
- Gain bandwidth product of FET is small than BJT.

Classification of FET

- FET
  - Junction field effect transistors (JFET)
  - Metal oxide semiconductor field effect transistor (MOSFET)
- Based on the construction JFETs are of two types
  - N Channel FET
  - P Channel FET

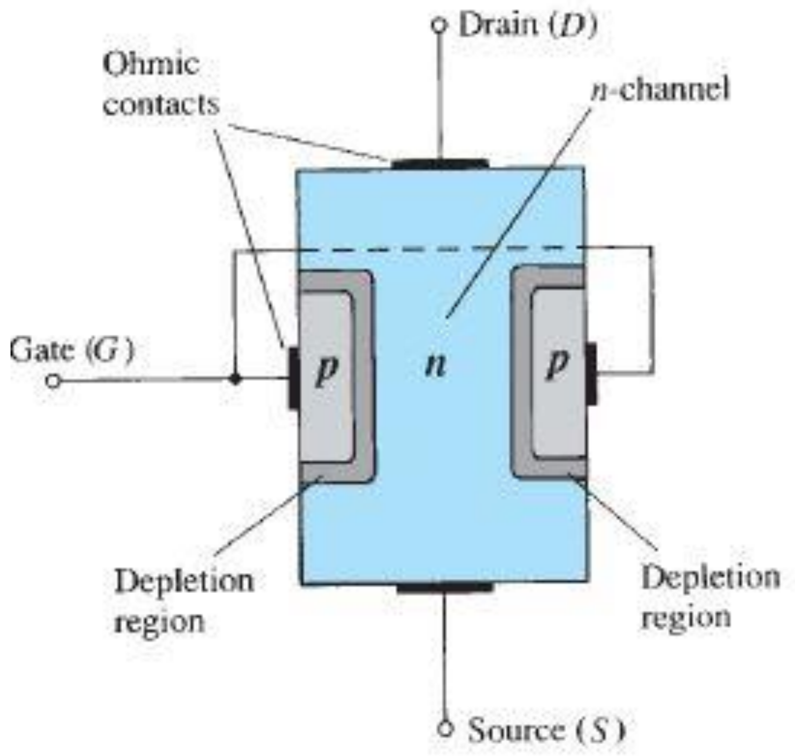


Terminals of JFET

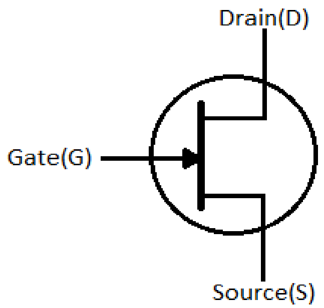
Source: The source is the terminal through which majority carriers enter the Silicon Bar

Drain: Terminal through which majority carriers leave the bar

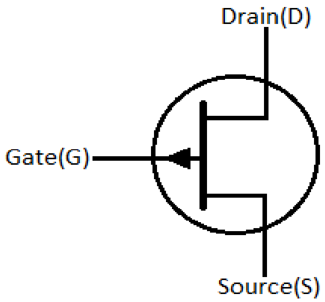
Gate: controls Drain current and is always reverse biased



Symbols of JFET:



N-channel JFET



P-channel JFET

Classification of FET

- The operation of FET can be compared to the water flow through a flexible pipe
  - When one end is pressed the cross sectional area decreases hence water flow decreases
  - In a FET drain is similar to outlet.

Water analogy for the JFET control mechanism.

Difference between N and P channel JFET

N Channel FET	P Channel FET
1. Current carriers are Electrons	1.Current carriers are holes
2. Mobility of electrons is almost twice that of Holes in P channel FET	2.Mobility of holes is poor
3. Low input Noise	3.More noise
4. Large Transconductance	4.Low Transconductance

Operation of JFETs

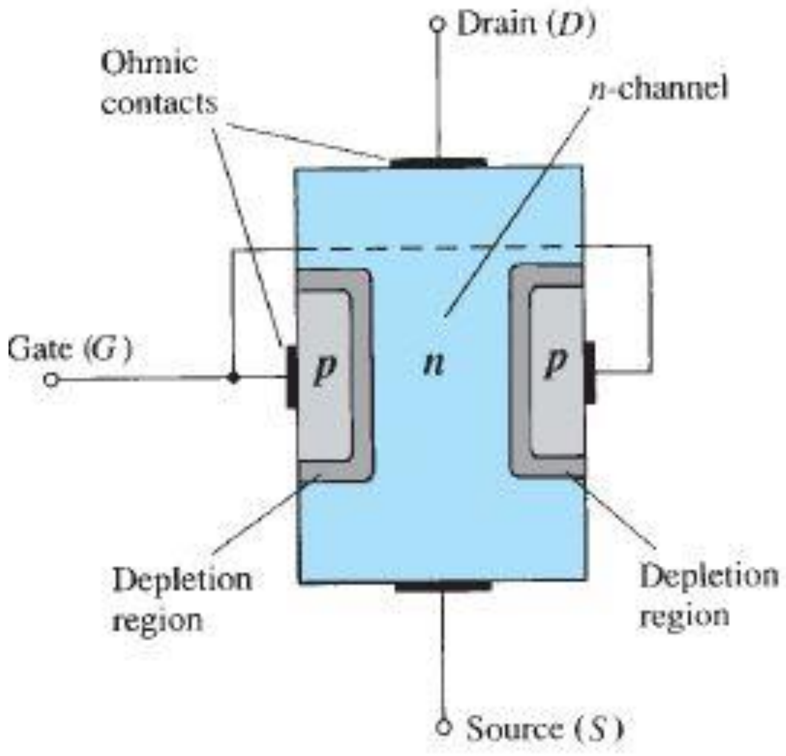


Fig. JFET at V_{GS} = 0 V and V_{DS} = 0 V.

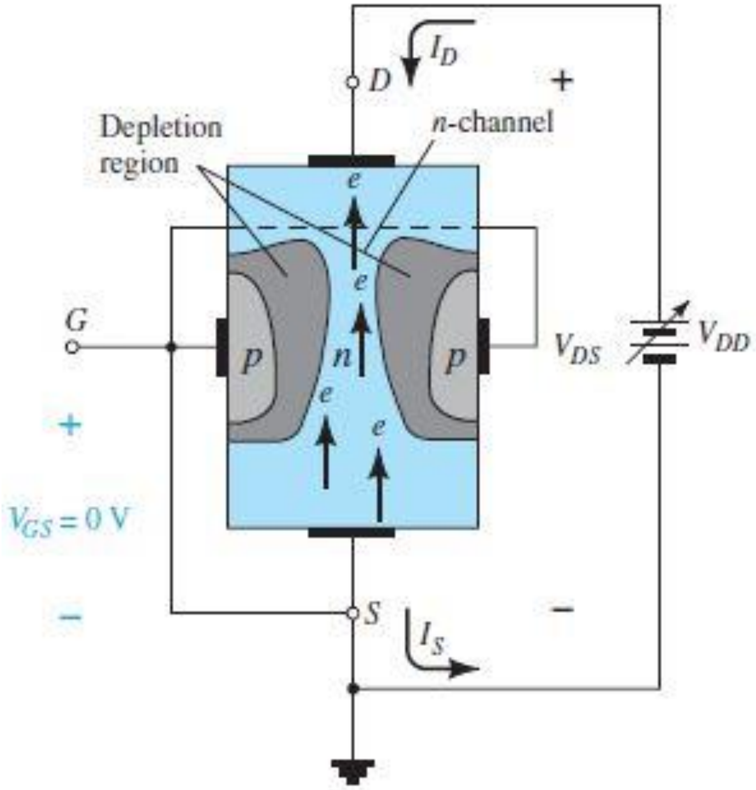


Fig. JFET at V_{GS} = 0 V and V_{DS} >0 V.

Operation of JFETs

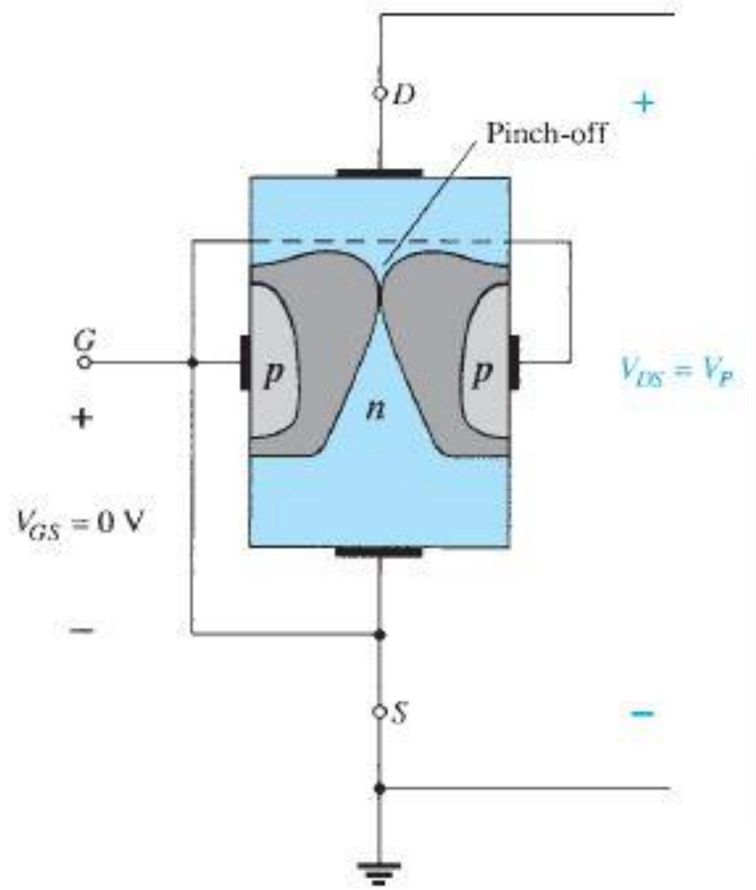


Fig. Pinch off voltage V_{GS} = 0 V and V_{DS} =V_P.

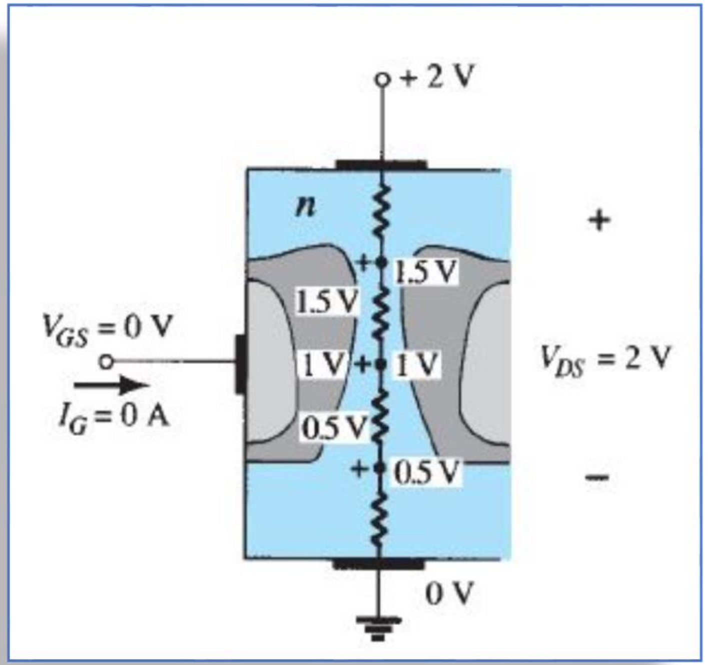


Fig. Varying reverse-bias potentials across the p–n junction of an n-channel JFET.

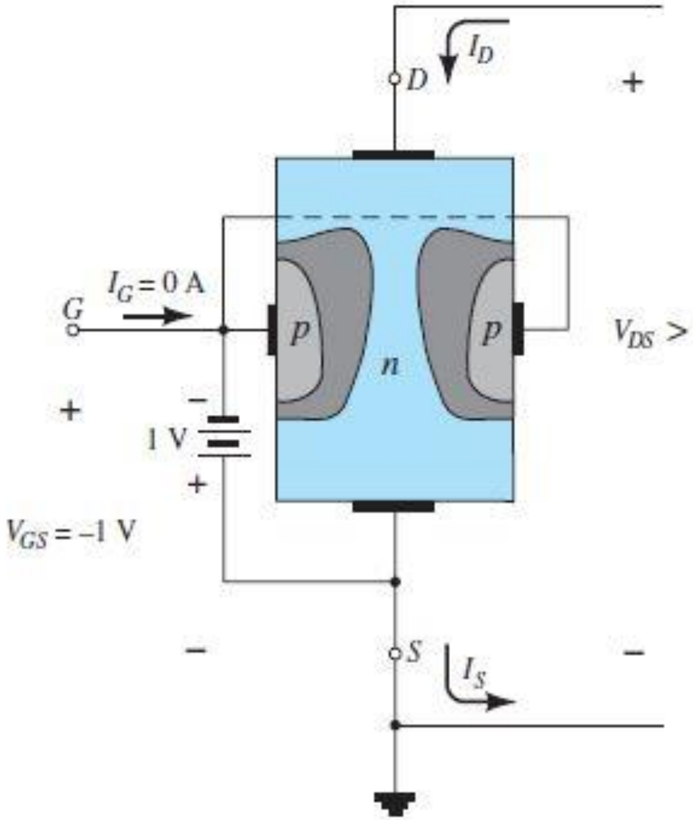
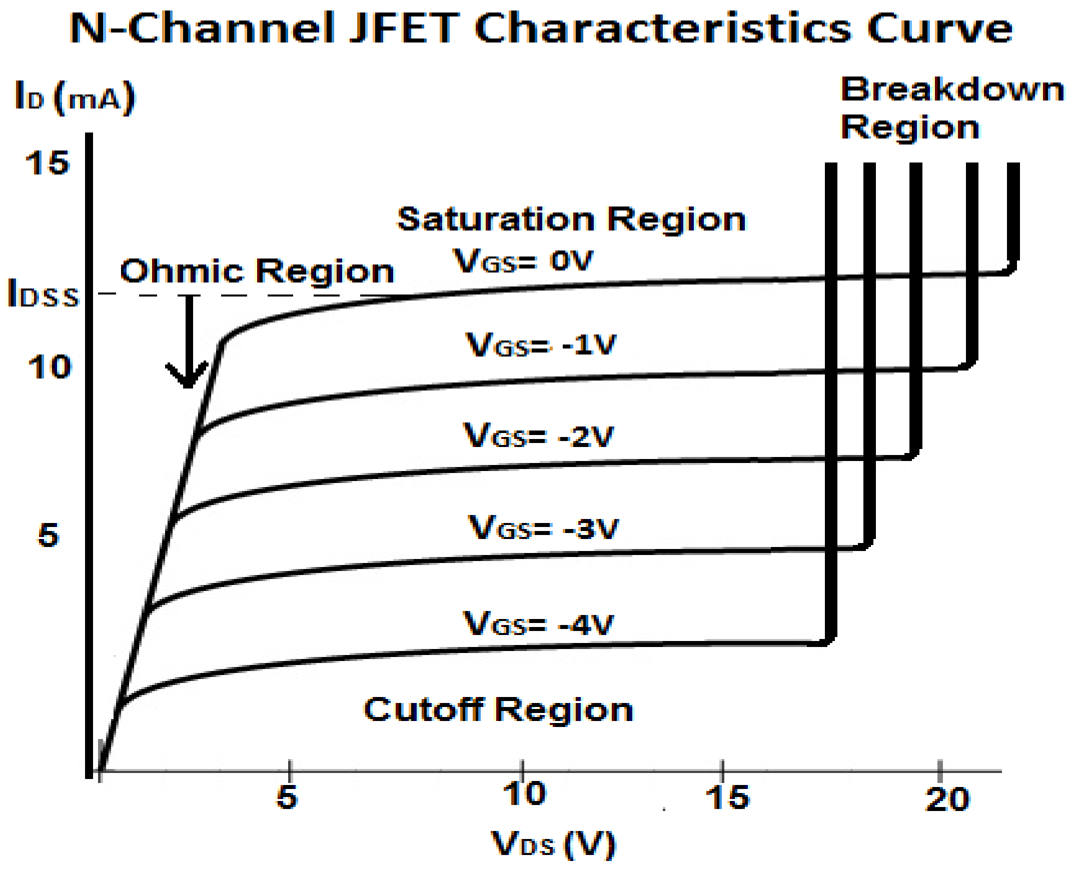


Fig. Pinch off V_{GS} = negative voltage

JFET characteristics

1. **Output or drain characteristics**: Curve between drain current and drain to source voltage for different value of gate to source voltage.



JFET characteristics

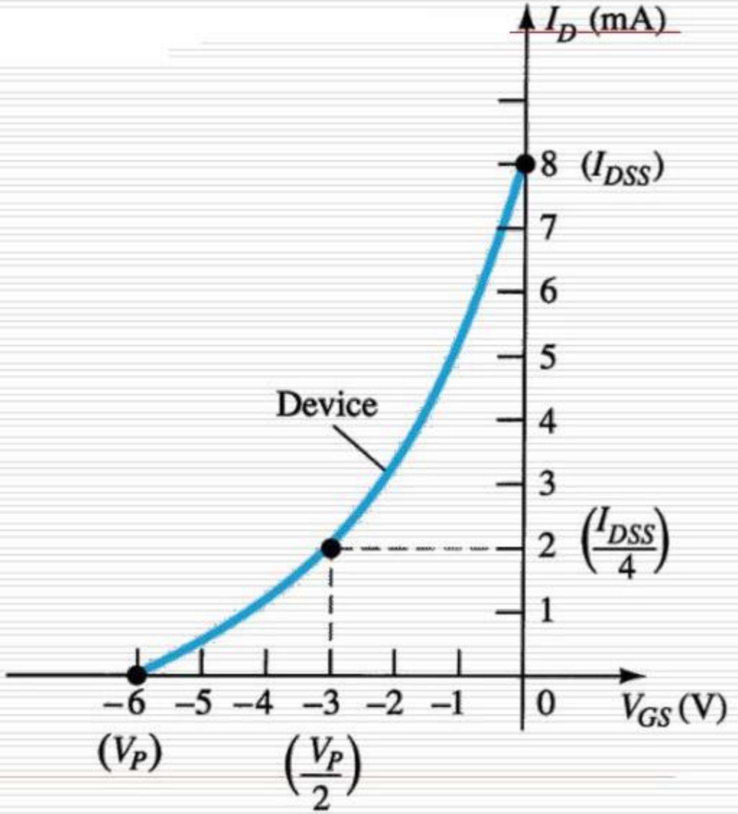
1. Transfer characteristics

control variable

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

constants

Vgs	I_D
0	I_{DSS}
0.3Vp	$I_{DSS}/2$
0.5Vp	$I_{DSS}/4$
Vp	0 mA



Thank you